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Watching alone: Relational goods, television and happiness

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Abstract

This paper investigates the role of relational goods for subjective well-being. Using a large sample of individuals from the World Values Survey, we find that relational goods have a significant effect on life satisfaction, while television viewing plays a key role in crowding-out relationality. Both results are robust to the use of alternative indicators of relationality and to instrumental variable estimation to deal with possible simultaneity. The findings suggest that the relational treadmill can provide an additional explanation of the income–happiness paradox: the effect of higher income on happiness is offset by lower consumption of relational goods, with television playing a significant role in explaining underconsumption of relationality.
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1. Introduction

Happiness is back in economics. After a long absence, the theoretical debate about the relationship between economics and happiness is currently growing steadily. In the early 1970s, Easterlin (1974), influenced by developments in psychology (Brickman and Campbell, 1971), opened up the debate on the income–happiness paradox with an important empirical finding: in 30 surveys over 25 years, per capita real income rose by more than 60 percent, but the proportion

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of people who rated themselves as "very happy", "fairly happy" or "not too happy" remained almost unchanged.[2]

Among the many explanations that have been offered for the income–happiness paradox, one of the most popular among economists is based on the relative (or positional) consumption hypothesis. The basic idea is that people compare themselves to some reference group when making consumption decisions, so that individual utility depends not only on the *absolute* level but also on the *relative* level of consumption (Frank, 2005; Layard, 2005). As a consequence, if everyone becomes richer and enjoys higher consumption, nothing changes in relative terms and individual happiness is unaffected.

It is important to observe that the positional theories commonly used to explain the income–happiness paradox rely on a particular idea of sociality: these theories are *social*, in the sense that positional competition cannot exist in the Robinson's island, but the kind of sociality taken into account has no reference to the *relational nature* of happiness. The main explanations of the income–happiness paradox do not consider interpersonal relationships as a source of happiness *per se*.[3]

In psychology, interpersonal relationships have long been recognized, both theoretically and empirically, as a key determinant of well-being.[4] The relational dimension is so important that some theorists have defined *relatedness* as a basic human need that is essential for well-being (Baumeister and Leary, 1995; Deci and Ryan, 1991). In particular, within the *eudaimonic* approach, many authors see a universal association between the quality of relationships and well-being (see, e.g. Argyle, 1987; Myers, 1999; Deci and Ryan, 2001; Diener and Seligman, 2002). Research on intimacy also highlights the importance of relatedness for well-being and underlines that it is the quality of relatedness that determines well-being.[5]

Although the analysis of the social dimension of economic behaviour has an old tradition, economics started only recently to explore the effects of the quality of interpersonal relationships on economic performance, welfare and happiness (see, e.g. Gui and Sugden, 2005). Mainstream economics does not deal with the *direct* link between sociality intended as relationality and individual well-being.[6] What economics may need, in our view, is a better understanding of the links between non-instrumental social interaction and subjective well-being.

[2] Easterlin also found that within a single country, at a given point in time, the correlation between income and happiness exists and is robust. Across countries, instead, poorer countries do not always appear to be less happy than richer countries (Easterlin, 1974, p. 100).

[3] Neumark and Postlewaite (1998), for example, show that if a woman's husband earns less than her sister's husband, the first woman is more likely to go out to work in order to keep up with the living standards of her sister. This, however, is different from considering family relations as a direct source of happiness.

[4] Bradburn (1969), one of the classic works on well-being, finds that social relationships are one of the strongest correlates of positive emotions. Baumeister and Leary (1995) review the evidence showing that people have a fundamental need for close social relationships. A recent study by Kahneman et al. (2004) finds that affect balance (positive minus negative emotions) is greater when people are alone rather than with others in only 1 out of 15 daily activities.

[5] Nezlek (2000) reviews a number of studies showing that the quality of relationships, as opposed to the quantity, helps to predict well-being. Carstensen (1995) and Kasser and Ryan (1999) point to the same conclusion. Ryff and Singer (2000) illustrate theoretically and empirically the links between interpersonal relationships, physiological functioning and happiness.

[6] The recent work by Meier and Stutzer (2004) represents a major exception. Using the German Socio-Economic Panel for the period between 1985 and 1999, this study finds robust evidence that volunteers are more satisfied with their lives than non-volunteers.

This paper moves along this direction of research. We start by discussing the concept of relational goods, clarifying how it relates to the existing literature on the social dimension of economic behaviour, and emphasizing its focus on the dimensions of *genuineness* and *identity* of interpersonal relationships. We then present a detailed empirical investigation of the effects of relational goods on life satisfaction. We also examine the role of television in explaining why individuals underconsume relational goods; following Putnam (1995), who identified television as a key factor in reducing social capital, we argue that television, because of its intrinsic characteristics, plays a key role in crowding-out interpersonal relationships.

Using individual data from the World Values Survey, we find a positive effect of indicators of relational goods on life satisfaction, and a negative impact of television viewing on relational activities, controlling for individual demographic and socio-economic factors, personality characteristics and environmental differences. Both empirical relationships are strongly significant and robust to the use of alternative indicators of relationality and to instrumental variable estimation to deal with possible simultaneity. We interpret these findings as an indication that lower consumption of relational goods may contribute to offset the impact of higher income levels, thus providing an additional explanation for the income–happiness paradox. The crowding-out effect of television viewing is indicated as an explanation of individual underconsumption of relational goods.

The paper is structured as follows. Section 2 discusses the concept of relational goods and its role in the debate on happiness and economics. Section 3 illustrates the direct and indirect effects of television viewing on relationality. Section 4 discusses the methodology and the data set used for the empirical analysis, while the results are presented in Section 5. Section 6 concludes with a discussion of the main implications of the analysis.

2. Relational goods

The analysis of interpersonal relations within economics has old origins. In the *Theory of Moral Sentiments*, Smith (1959) identified non-instrumental relationality, intended as fellow-feeling or sympathy, as an essential element for human happiness. Thereafter, however, the *relational* dimension of economic behaviour was largely neglected. The *social* dimension of economic behaviour was central in the thought of Veblen (1899), whose concept of conspicuous consumption was later developed by Duesenberry (1949) in his theory of relative consumption. More recently, Becker (1973, 1974) placed interpersonal relationships at the core of his theoretical analysis of human behaviour. Within the same methodological line, the New Social Economics explores non-contractual influences on other agents' preferences, information and constraints (e.g. Blume and Durlauf, 2001).[7]

While these theories focus on the social dimension of economic behaviour, they place relatively less emphasis on the relational content of interpersonal interactions. In the last two decades, however, behavioural and experimental economics have offered important contributions indicating that interpersonal dimensions, such as trust, fairness and reciprocity, are relevant for economic decisions (see, e.g. Bowles and Gintis, 2004; Fehr and Schmidt, 2006). In a line of thought that is closely connected to Smith's *Theory of Moral Sentiments*, other dimensions of interpersonal

[7] See Bruni (2004, 2005) for details on the historical evolution of the role of interpersonal relationships in economic theory.

relationships have recently entered the theoretical debate on social interactions under the label *relational goods*.[8]

Uhlaner (1989), one of the first authors to use this concept, together with Gui (1987), defined relational goods as goods that "can only be 'possessed' by mutual agreement that they exist, after appropriate joint actions have been taken by a person and non-arbitrary others" (1989, p. 254). Relational goods cannot be produced, consumed, or acquired by a single individual, because they depend on the interaction with others and are enjoyed only if shared with others.[9] According to Uhlaner, "goods which arise in exchanges where anyone could anonymously supply one or both sides of the bargain are not relational" (p. 255). A first key feature of relational goods, therefore, is that *identity matters*.

More recently, Gui and Sugden (p. 3) defined relational goods as "the affective components of interpersonal relations [that] are usually perceived as having value through their sincerity or genuineness". In the words of Nussbaum (1986, ch. 12), "it is the relationship itself that constitutes the good", as opposed to the exchange of goods and services that may occur in the same interaction. The non-instrumental nature, or *genuineness*, of the relationship can therefore be considered the second essential feature of relational goods.[10]

Summing up, while both in psychology and in mainstream economics the terms sociality and interpersonal relationships are generally broadly defined, economists who have recently adopted the concept of relational goods use it in a narrower sense. The dimension of genuineness is foundational, and the identity of the other person is essential for the value, and in some cases even for the existence, of the relational good.

In the empirical analysis presented in this work, we use as proxies of relational goods two separate sets of indicators that closely embody the features of identity and genuineness that characterize relational goods. The first set refers to active involvement in individual volunteering activities (church-religious, sport-recreation, art-educational, labor union, political party, environmental, professional, charity), thus focusing more directly on the genuineness component of relational goods. The second set includes time spent with specific groups of people (relatives, friends, colleagues, people at church, people at service organization), thus focusing more explicitly on the identity component of relational goods. It is important to observe that the comparative analysis of individual relational indicators that differ in terms of both the identity and genuineness components allows us to capture more closely the specific features of relational goods and, therefore, to provide a more accurate assessment of their importance for subjective well-being.

3. Television and relational goods

Watching television is the most important leisure activity worldwide. According to recent survey data (IP, 2004), the average daily television viewing time for adults is 217 min in Western

[8] As Gui and Sugden (2005, p. 2) put it, "what we do not find sufficiently stressed are some peculiar elements that come into play when interactions are highly personalised".

[9] Relational goods share some characteristics of local public goods: there is no rivalry in consumption, although the contribution to the good may be asymmetrical. Indeed, relational goods require more than mere non-rivalry. For a discussion of this point, see Gui (2002, p. 45).

[10] A typical relational good identified by Borcherding and Filson (2002) is the "companionship" derived from eating lunch with one's colleagues instead of eating alone, a good which is distinct from the exchange of goods and services possibly occurring in the same interaction.

Europe, 228 min in Eastern Europe and 290 min in the United States. These figures imply that by the age of 75 an average European has spent about 12 full years watching television out of approximately 50 waking years, assuming an average of 8 h of sleep per day. Not only are viewing time levels very high, but also the number of hours spent watching television continues to grow steadily. Between 1995 and 2003, the average daily viewing time for adults in Western Europe has risen from 196 to 217 min.[11]

Given the prominent and increasing role played by television in people's life, its influence on individual behaviour and well-being cannot be ignored. Most of the extensive literature on the effects of television focuses on the sociological and psychological dimensions: the impact of television content on social perceptions, attitudes and beliefs, or the effects of television viewing on individuals' mental processes and health outcomes.

Economists have focused on the impact of television advertising on consumer behaviour, but otherwise have largely ignored the effects of television on economic decisions.[12] Quite surprisingly, there have been hardly any studies examining the effects of television on individual happiness from an economic perspective. The recent paper by Frey et al. (2005) represents an important exception. In their study, high levels of television consumption are found to be negatively related to individual life satisfaction and are interpreted as the result of imperfect self-control and mispredicted utility. In our analysis, we focus on one specific channel through which television can affect well-being. We argue that television viewing has a negative impact on individual happiness by harming, and to some extent replacing, relationships with other people. Because of its specific characteristics, television plays an important role in crowding out relational activities both directly and indirectly.

First, consider the direct effect. The time spent watching television is generally subtracted from communicating with family and friends, participating to community-life or interacting socially, that is, relational activities that contribute significantly to our life satisfaction. There is extensive evidence that television viewing has a profound impact on relationships within the family, leading in particular to less communication and interaction.[13] Television viewing has also been shown to decrease the amount of time spent with friends (e.g. Robinson, 1977): "People no longer sit around and visit. Everywhere you go you have to outtalk TV. TV people have entered your home and life more than people who should be friends and companions" (Kubey and Csikszentmihalyi, 1990, p. 108).

Second, the indirect effect. Television is one of the main agents of socialization and, through advertising and program contents, plays a key role in defining what our goals should be. In particular, television is a key factor in producing the belief that happiness depends on material consumption and in raising our material aspirations. When watching television, people are overwhelmed by images of more and better products than what they have, and people richer and wealthier than they are. Both factors contribute significantly to raise material aspirations.[14] Television thus leads

[11] The average daily viewing time for adults increased from 208 to 228 min between 2000 and 2003 in Eastern Europe, and from 283 to 290 min between 2001 and 2003 in the United States (IP, 2004).

[12] The work by Corneo (2005) on the relationship between television viewing and working time represents a major exception.

[13] "Television seems to have changed the ways in which family interaction occurs. When the set is on, there is less conversation and less interaction. [...] There is more privatization of experience; the family may gather around the set, but they remain isolated in their attention to it". [...] "When the TV set is on, it freezes everybody. Everything that used to go on between people – the games, the arguments, the emotional scenes, out of which personality and ability develop – is stopped" (Kubey and Csikszentmihalyi, 1990, p. 108).

[14] See Bruni and Stanca (2006) for an analysis of the impact of television viewing on material aspirations.

individuals to underestimate the relative importance of relational goods for their life satisfaction and, as a consequence, to over-invest in income-producing activities and under-invest in relational activities.[15]

A natural question then follows. If television viewing is so damaging to relational activities and, as a consequence, to individual happiness, why do people choose to watch so much TV? In our view, part of the answer lies in the fact that relational activities are constrained by immediate costs in terms of time and effort, the necessity of other people to participate, and the need to engage intensively for long periods. In contrast, television has a number of features that make it an easily accessible alternative. First, the mentally undemanding and physically passive nature of television viewing contributes to making it a very attractive relaxing activity. Second, television is extremely inexpensive and available at any time of the day. Third, television tends to produce a viewing habit and, in many cases, can become highly addictive (see, e.g. Winn, 1985).

But the most important reason for the crowding-out effect on relational goods lies in television's own nature: television provides, at virtually no costs and without any effort, not only a relaxing and convenient entertainment, but also a *virtual network of relationships and interactions* that, despite being completely artificial and illusory, tend to become a substitute for actual social relationships. People tend to watch talk-shows and reality-shows as if these were part of their own social life and to talk about TV people as if they had an actual relationship with them. In short, television provides inexpensive and effortless artificial relationality that is commonly and increasingly used as a substitute for interpersonal relationships.

4. Data and methodology

Based on the discussion in the previous sections, we investigate two main propositions using individual data, controlling for relevant demographic, socio-economic, personality and environmental characteristics: first, relational goods have a positive independent effect on life satisfaction; second, television viewing has a crowding-out effect on relational activities. This section briefly presents the data set and the methodology used for the empirical analysis.

We use individual data from the World Values Survey (WVS), a compilation of surveys conducted in more than 80 countries representing about 85 percent of the world's population (see Inglehart, 2000, 2004).[16] Four WVS waves are currently available (1980–1982, 1990–1991, 1995–1997 and 1999–2001), for a total of 264,778 observations. The sample size in our analysis is smaller, however, due to the limited availability of individual variables. In particular, only the fourth wave contains information about relational time, and information on voluntary activities is not available in the first wave. Information on television consumption is only available for the third and fourth waves, for a smaller set of countries. Summary statistics for all the variables used in the analysis are reported in Table 1.

In order to study the first proposition, we assume that individual happiness (HAP) depends on demographic factors (DEMO), socio-economic conditions (SEC), personality traits (PERS),

[15] A similar point is made by Frey and Stutzer (2004), who argue that individuals tend to underestimate the utility of personal interactions.

[16] The WVS provide information on individual beliefs about politics, the economy, religious, social and ethical topics, personal finances, familial and social relationships, happiness and life satisfaction. Within each country, samples are selected randomly "from all administrative regional units after stratification by region and degree of urbanization" (Inglehart, 2000, p. 7).

Table 1
Descriptive statistics

Variable	Mean	S.D.	Min	Max	No. of observations
Time spent with: family	3.48	0.81	1.00	4.00	53, 501
Time spent with: friends	3.27	0.89	1.00	4.00	94, 007
Time spent with: colleagues	2.48	1.19	1.00	4.00	77, 759
Time spent with: church	2.24	1.24	1.00	4.00	76, 920
Time spent with: sport	1.91	1.14	1.00	4.00	76, 472
Voluntary org. (number)	1.19	1.66	0.00	8.00	193, 344
Active vol. org. (number)	0.51	0.99	0.00	8.00	193, 344
Voluntary org. (dummy)	0.55	0.50	0.00	1.00	193, 344
Active vol. org. (dummy)	0.31	0.46	0.00	1.00	193, 344
Vol. organization: church	0.13	0.33	0.00	1.00	226, 894
Vol. organization: sport	0.10	0.30	0.00	1.00	200, 504
Vol. organization: arts	0.07	0.25	0.00	1.00	227, 264
Vol. organization: unions	0.04	0.19	0.00	1.00	229, 753
Vol. organization: politics	0.04	0.19	0.00	1.00	231, 050
Vol. organization: environment	0.03	0.17	0.00	1.00	229, 882
Vol. organization: professional	0.04	0.20	0.00	1.00	229, 875
Vol. organization: charity	0.06	0.23	0.00	1.00	228, 352
Life satisfaction	66.39	24.85	10.00	100.00	259, 947
Income decile	4.73	2.53	1.00	10.00	221, 148
Health	3.75	0.93	1.00	5.00	213, 305
Freedom	6.63	2.45	1.00	10.00	245, 836
Unemployed dummy	0.08	0.27	0.00	1.00	262, 882
Married dummy	0.63	0.48	0.00	1.00	260, 449
Education (years)	4.65	2.35	1.00	16.00	188, 968
Age	41.15	16.35	15.00	101.00	251, 323
Male dummy	0.48	0.50	0.00	1.00	260, 130
Trust dummy	0.30	0.46	0.00	1.00	252, 181
Honesty	8.58	2.36	1.00	10.00	246, 262
Religion important	2.89	1.08	1.00	4.00	227, 079
Friends important	3.26	0.73	1.00	4.00	229, 932
TV viewing	1.58	0.96	0.00	3.00	92, 049
TV viewing dummy	0.48	0.50	0.00	1.00	92, 049
TV important	2.59	0.91	1.00	4.00	36, 420
TV important dummy	0.55	0.50	0.00	1.00	36, 420

Note: See Section 4 for details on the definition and construction of the variables.

environmental characteristics (ENV) and consumption of relational goods (REL):

$$\mathrm{HAP}_i = f(\mathrm{DEMO}_i, \mathrm{SEC}_i, \mathrm{PERS}_i, \mathrm{ENV}_i, \mathrm{REL}_i). \tag{1}$$

Happiness is measured by life satisfaction, based on the question: "All things considered, how satisfied are you with your life as a whole these days?"[17] The set of control factors includes age, gender, income, health, freedom, education, employment status, marital status, personality traits and beliefs (trust, honesty, importance of religion).

Income is measured by self-reported deciles in the national distribution of income, so that income levels expressed in relative terms are comparable across countries and individuals. Health

[17] The original answers on a scale 1 (dissatisfied) to 10 (satisfied) were multiplied by 10 in order to ease interpretation of regression results. See, Frey and Stutzer (2002) for a discussion of the use of reported subjective well-being as an empirical approximation of individual happiness.

is measured by the self-assessed state of health, on a 1–5 scale (very good = 5, good = 4, fair = 3, poor = 2, very poor = 1). Freedom is defined as the degree of freedom of choice and control an individual has over his life, on a 1–10 scale (1 = none at all, 10 = a great deal). The honesty indicator is based on answers to the question, "Is it justifiable to cheat on taxes?", on a 1–10 scale (1 = always justifiable, 10 = never justifiable). The trust dummy takes the value 1 for those who think that "in general people can be trusted" (0 if "you cannot be too careful when dealing with people"). The indicator of importance of religion is measured on a 1–4 scale (1 = not very important, 4 = very important).

Relational activities are measured by two sets of indicators. The first set refers to volunteering activities. In particular, we rely on information on both membership and active involvement in the following voluntary activities: (a) church-religious, (b) sport-recreation, (c) art-music-educational, (d) labor union, (e) political party, (f) environmental, (g) professional and (h) charity.[18] It is important to observe that the information available in the WVS allows us to identify the net effect of the relational component of voluntary activities (actual involvement, as opposed to simple membership), and also to assess and compare the role played by individual voluntary activities.[19] Fig. 1 shows the frequency of either membership (left panel) or active participation (right panel) to individual voluntary organizations. Fig. 2 displays the relation between active participation in individual voluntary organizations and life satisfaction.

The second set of indicators refers to "relational time", that is time spent with (a) parents or other relatives, (b) friends, (c) colleagues from work or profession, (d) people at church and (e) people at service organization.[20] Figs. 3 and 4 present the frequency distribution for each indicator of relational time and their relation with life satisfaction, respectively.

In order to study the second proposition, we assume that consumption of relationality depends on demographic factors, socio-economic conditions, personality traits, environmental characteristics and on the level of television viewing:

$$\mathrm{REL}_i = f(\mathrm{DEMO}_i, \mathrm{SEC}_i, \mathrm{PERS}_i, \mathrm{ENV}_i, \mathrm{TV}_i). \tag{2}$$

Television consumption is measured from answers to the question, "Do you ever watch television? If yes, how much time do you usually spend watching television on an average weekday (not weekends)?"[21] From this variable, we constructed a dummy variable equal to 1 for high-TV viewers (more than 2 h per day) and 0 for low-TV viewers (no TV or 1–2 h per day). This definition splits the sample almost equally (the sample mean of the TV dummy is 0.48). We also constructed a second dummy variable for heavy TV viewers (more than 3 h per day). Fig. 5 displays the frequency of television viewing, while Figs. 6 and 7 present the relation between television viewing and active volunteering activities and relational time indicators, respectively.

In order to control for cultural and societal differences that may play a key role in explaining international differences in subjective well-being (Diener, 2000), all the equations include individual country dummies, so that unobserved heterogeneity due to country-level environmental

[18] This information is based on the questions, "To which of the following voluntary organizations, if any, do you belong to?" and "For which of the following voluntary organizations, if any, are you currently doing unpaid voluntary work?".

[19] These two dimensions are not available, for instance, in the panel data set used by Meier and Stutzer for their study of the effect of volunteering on life satisfaction.

[20] These variables are constructed on the basis of answers to the question, "How often do you spend time with" (1 = not at all, 2 = only a few times a year, 3 = once or twice a month, 4 = weekly). Given the ordinal nature of the original variables, for each indicator we also constructed a dummy variable taking value 1 if once a month or weekly, 0 otherwise.

[21] The options presented to the respondents are (1) "Do not watch TV or do not have access to TV"; (2) "1–2 h per day"; (3) "2–3 h per day"; (4) "More than 3 h per day".

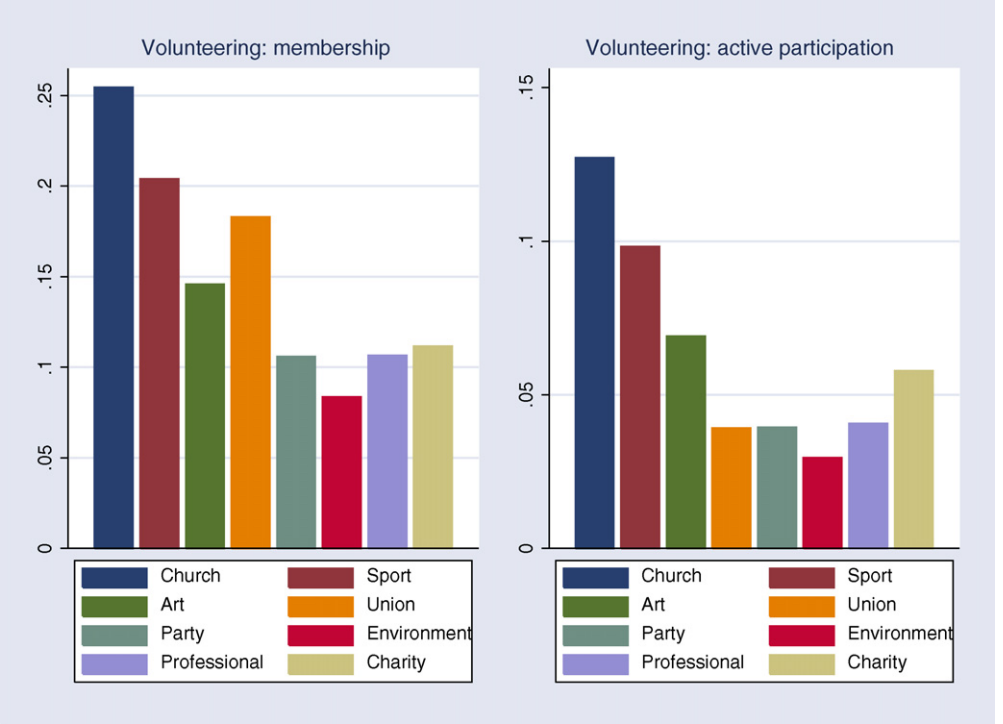

Fig. 1. Frequency of individual volunteering activities.

differences is controlled for. The set of regressors also includes time fixed effects to allow for heterogeneity between different survey waves. A linear functional form is assumed, and the different specifications of Eq. (1) are estimated by OLS.[22] Eq. (2) is estimated either by OLS or using a probit model when the dependent variable is a dummy variable. Test statistics are calculated using heteroskedasticity robust standard errors.

The causal interpretation of regression results is generally difficult in social sciences, but it is a particularly delicate issue in the present analysis, given the characteristics of the relationships under investigation and the cross-sectional nature of the data set. The finding of an association between relational activities and life satisfaction, or between television viewing and relational activities does not by itself imply a causal relationship.[23] In fact, in both these relationships there might be a reverse causality: in the first case, life satisfaction could determine higher levels of relational activities and in the second case, high levels of relationality could lead to lower consumption of television. Surely, it cannot be ruled out that causality goes in both directions. The possibility that endogeneity may result from omitted variables affecting both dependent

[22] Qualitatively similar results are obtained using ordered probit estimators, which would more appropriately take into account the ordinal nature of the dependent variable. OLS results are presented for ease of interpretation.

[23] Diener and Seligman (2004, p. 20) stress the causality problem in the correlation between social relations and well-being, concluding that "within-person data indicate that it is not just that happy people have better relationships than unhappy people. Rather, the same individuals are happier on average when they are with others than when they are alone. These results again suggest that positive social contact causes well-being. [. . .] Thus, the causal path from social relationships to well-being moves in both directions."

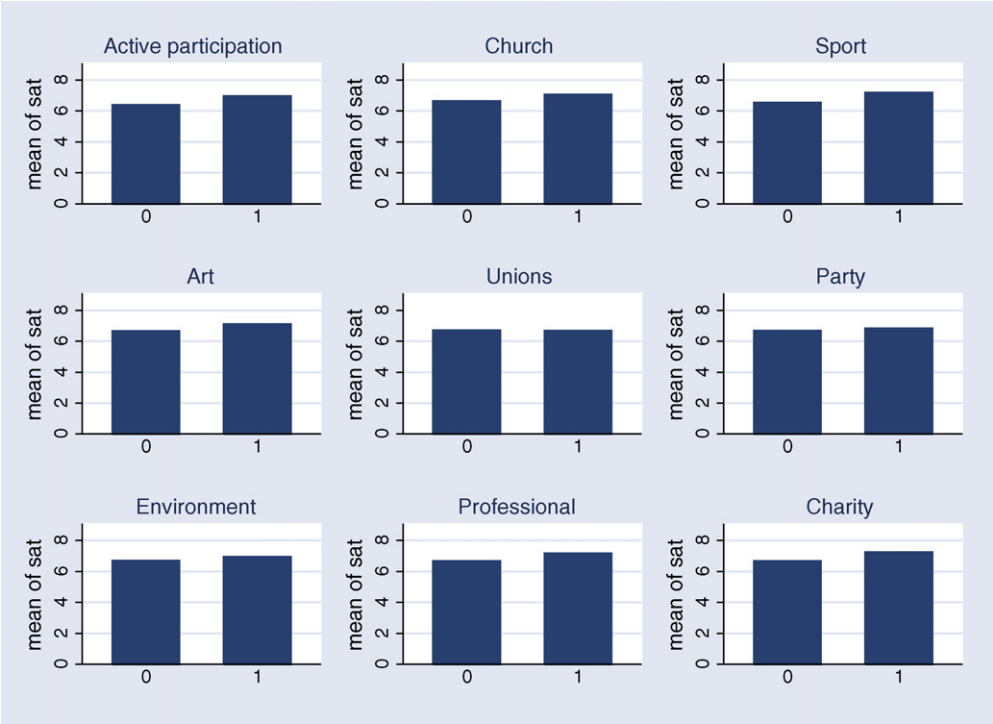

Fig. 2. Volunteering activities and life satisfaction.

and independent variables should also be considered (for example, extroversion determining both happiness and relational activities). However, the use of a wide range of individual control factors and country-specific dummy variables in estimating Eqs. (1) and (2) makes this cause of endogeneity a relatively minor concern.

In order to check the causal interpretation of the results, we also estimate the relationships of interest by instrumental variables (2SLS). Although it is generally quite difficult to find appropriate instruments, we use the variables "importance of friends" and "importance of family" as instruments for the relational indicators in Eq. (1), under the assumption that the priority assigned to friends and family is related to the actual consumption of relationality, but is not independently related to life satisfaction.[24] For Eq. (2), we use the variable "importance of television" as an instrument for TV consumption under the assumption that self-reported television importance is related to levels of television viewing but is not independently related to relational goods.[25]

[24] The variables "importance of friends" and "importance of family" are based on the question, "Indicate how important friends (or family) are in your life" (1 = not at all, 2 = not very important, 3 = rather important, 4 = very important). We also constructed corresponding dummy variables (1 if rather or very important, 0 otherwise) and checked the robustness of the results.

[25] The indicator of television importance is constructed from the question, "Do you agree with the following statement: Television is my most important form of entertainment (1 = disagree strongly, 2 = disagree, 3 = agree, 4 = agree strongly). We also constructed a corresponding dummy variable (1 if agree or agree strongly, 0 otherwise) and examined both variables as possible instruments.

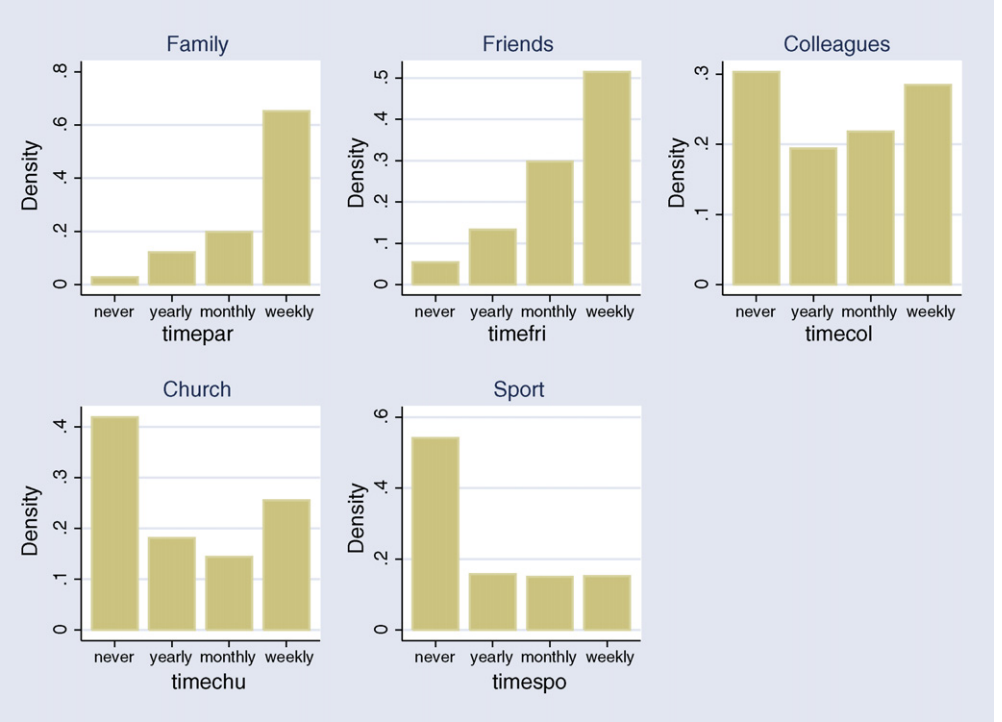

Fig. 3. Frequency of relational activities.

5. Results

5.1. Relational goods and life satisfaction

Table 2 presents OLS estimation results for Eq. (1), measuring relational goods with participation to voluntary organizations. The dependent variable is life satisfaction (measured on a scale from 10 to 100), and participation to voluntary organizations is measured by a dummy variable for either simple membership (column (a)), active participation (column (b)), or both indicators simultaneously (column (c)), controlling for the socio-economic and demographic variables described above. The estimates are based on observations for about 73,000 individuals, and all specifications include country-specific dummies and time-fixed effects for survey waves.[26]

We start by considering the results for the control variables in order to provide a preliminary assessment of the empirical specification. Across individuals from a wide range of countries, higher relative income is associated with higher subjective well-being, although the effect is relatively small: moving up by one decile in the relative income scale is associated with a strongly

[26] The sample includes WVS waves 2–4, with the following countries: Germany, Denmark, Spain, United States, Canada, Japan, Mexico, South Africa, Australia, Norway, Sweden, Argentina, Finland, South Korea, Switzerland, Puerto Rico, Brazil, Nigeria, Chile, Belarus, India, China, Taiwan, Austria, Lithuania, Latvia, Estonia, Ukraine, Russia, Peru, Venezuela, Uruguay, Philippines, Moldova, Georgia, Azerbaijan, Dominican Republic, Bangladesh, Serbia, Montenegro, Macedonia, Croatia, Bosnia-Hercegovina, Albania, Morocco, Vietnam, Zimbabwe and Uganda.

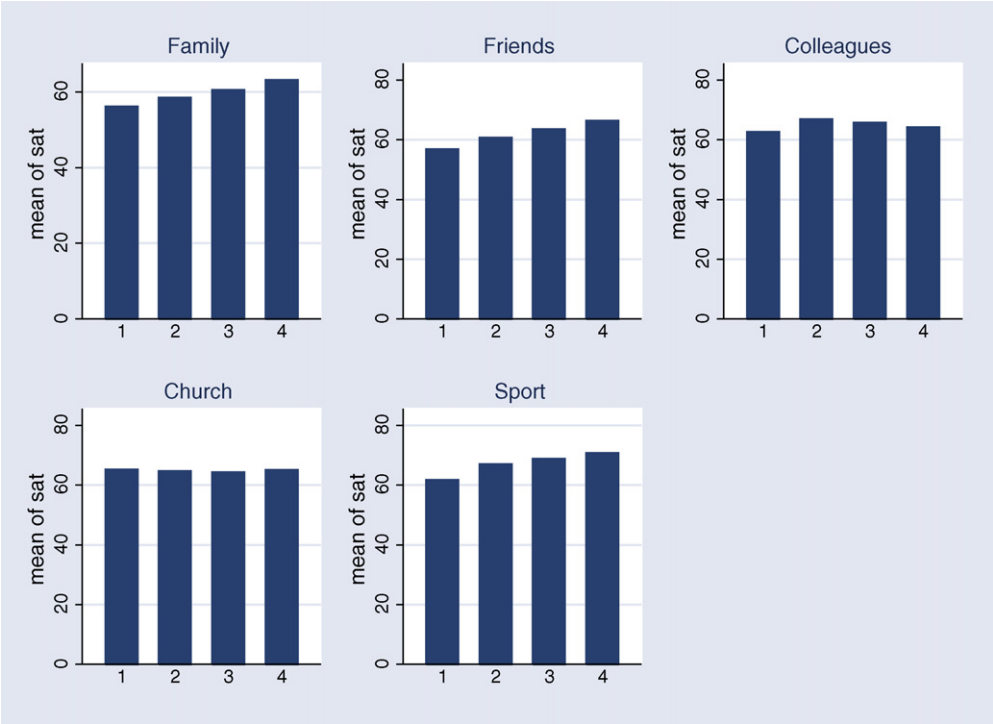

Fig. 4. Relational activities and life satisfaction.

significant 1.15 increase in life satisfaction (on a scale 10–100). The health and freedom indicators have large and highly significant positive coefficients.[27] Unemployment and marriage are associated with large and significant differences in life satisfaction (−3.9 and 3.67, respectively). The number of years of education has a positive but not significant effect on individuals' well being, with decreasing returns. Age is negatively and significantly related to life satisfaction. As for personality characteristics, individuals who assign a high priority to religion or think that cheating on taxes is never justifiable report systematically higher satisfaction levels. The same is true for those who think that, in general, people can be trusted.

Focusing on the relational indicators, membership in a voluntary organization is associated with a statistically significant increase in life satisfaction (1.15 percentage points), ceteris paribus. Active participation in the activities of a voluntary organization is also positively and significantly associated with higher life satisfaction, and the increase is somewhat larger and more significant. It is interesting to observe that the effect of volunteering on life satisfaction is as large as that of moving up by one decile in the income scale. When both volunteering indicators are considered jointly, so that the coefficient of the "active participation" dummy measures the net effect of the relational component (actual involvement), over and above simple membership, both variables have positive coefficients, and the effect of active participation is somewhat larger and more

[27] A one-point improvement in the self-assessment of health, on a five-point scale, is associated with a 4.8 increase in life satisfaction; a one-point improvement in freedom (defined as control over one's life), on a 10-point scale, is associated to a 3.12 increase in life satisfaction.

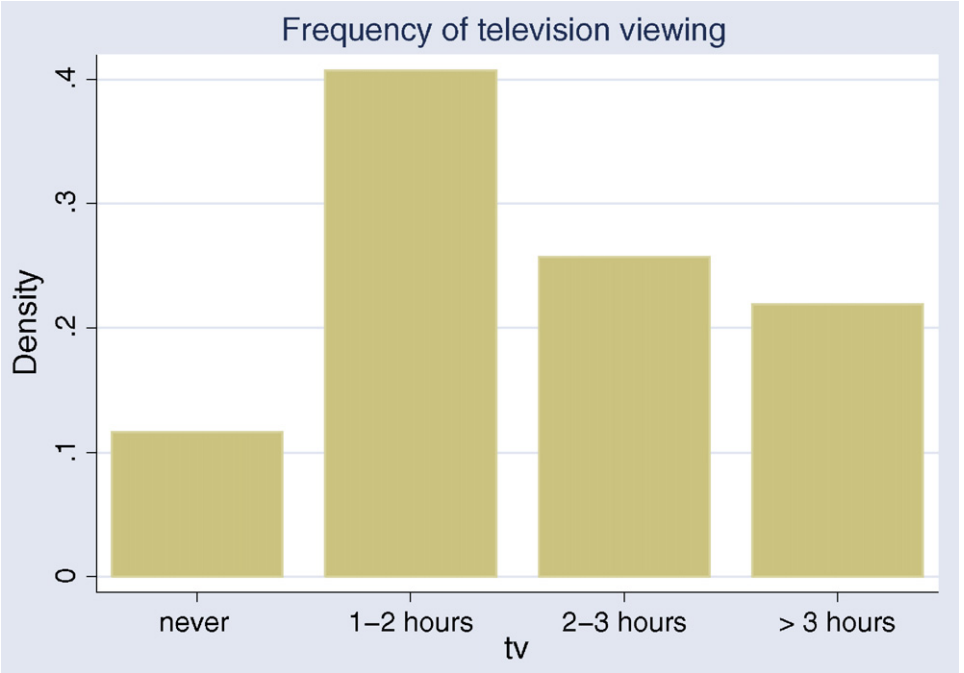

Fig. 5. Frequency of television viewing.

Table 2
Life satisfaction and participation in voluntary organizations

Independent variable	(a)	(b)	(c)
Vol. org. member (dummy)	1.15 (6.58)		0.73 (3.52)
Vol. org. active member (dummy)		1.17 (6.80)	0.75 (3.67)
Income	1.15 (32.07)	1.15 (32.10)	1.15 (31.99)
Health	4.81 (46.70)	4.81 (46.72)	4.80 (46.68)
Freedom	3.12 (77.89)	3.12 (77.87)	3.12 (77.83)
Unemployed	−3.90 (−13.34)	−3.94 (−13.48)	−3.91 (−13.35)
Married	3.67 (20.06)	3.67 (20.04)	3.68 (20.07)
Education	0.10 (0.73)	0.09 (0.72)	0.08 (0.63)
Education^2	−0.01 (−0.77)	−0.01 (−0.78)	−0.01 (−0.73)
Age	−0.59 (−20.13)	−0.59 (−20.03)	−0.59 (−20.13)
Age^2	0.01 (21.81)	0.01 (21.68)	0.01 (21.78)
Male	−1.10 (−7.10)	−1.10 (−7.10)	−1.11 (−7.20)
Religion important	1.01 (11.41)	0.99 (11.20)	0.98 (11.11)
Trust	0.89 (5.16)	0.89 (5.14)	0.88 (5.08)
Honesty	0.26 (7.44)	0.26 (7.43)	0.27 (7.45)
Adjusted R^2	0.35	0.35	0.35
Observations	73,572	73,572	73,572

Note: OLS estimates. Dependent variable: life satisfaction. *t*-Statistics in parentheses (heteroskedasticity robust standard errors). Regressors also include individual country dummies and time dummies for survey waves (1990–1991, 1995–1997, 1999–2001). Data source: World Values Survey 2–4 (Inglehart, 2000, 2004).

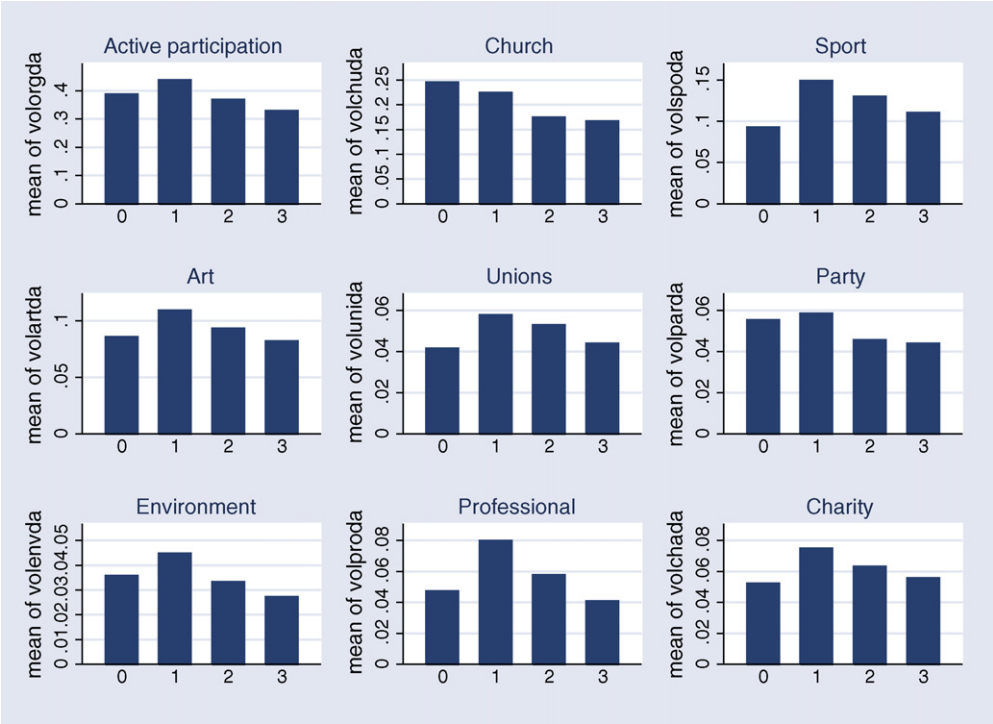

Fig. 6. TV viewing and volunteering activities.

strongly significant. These results suggest that the relational component of participation in voluntary organizations, represented by the actual interaction with other people, has an independent positive effect on life satisfaction.

Table 3 presents results obtained using the *number* of voluntary organizations to which people either simply belong or dedicate time actively (i.e. a discrete variable in a range from 0 to 8, as opposed to a binary variable). Membership in one additional voluntary organization is associated with a statistically significant 0.3 increase in life satisfaction, ceteris paribus. The number of active participations is also significantly associated to higher life satisfaction, and the increase is larger and more strongly significant. Interestingly, when both indicators are inserted jointly, both variables have positive coefficients, but only active participation is significant, whereas membership is no longer significant. These results confirm and qualify the earlier finding that the relational component of participation in voluntary organizations is the relevant factor for life satisfaction.

Table 4 presents results for the relationship between active participation in *individual* voluntary organizations and life satisfaction. The table reports results obtained by including the voluntary organization dummy variables either individually (column (a)) or jointly (column (b)). Participation in religious, sport, art and charity organizations has a positive and significant effect on life satisfaction in both specifications. In particular, participation in a charity voluntary organization has the largest effect on life satisfaction (1.78–1.58), whereas membership in sport organizations has the smallest effect (0.95–0.54). The coefficients for church- and art-related activities organizations are in the range 1.4–1.1 for the individual and joint specifications, respectively.

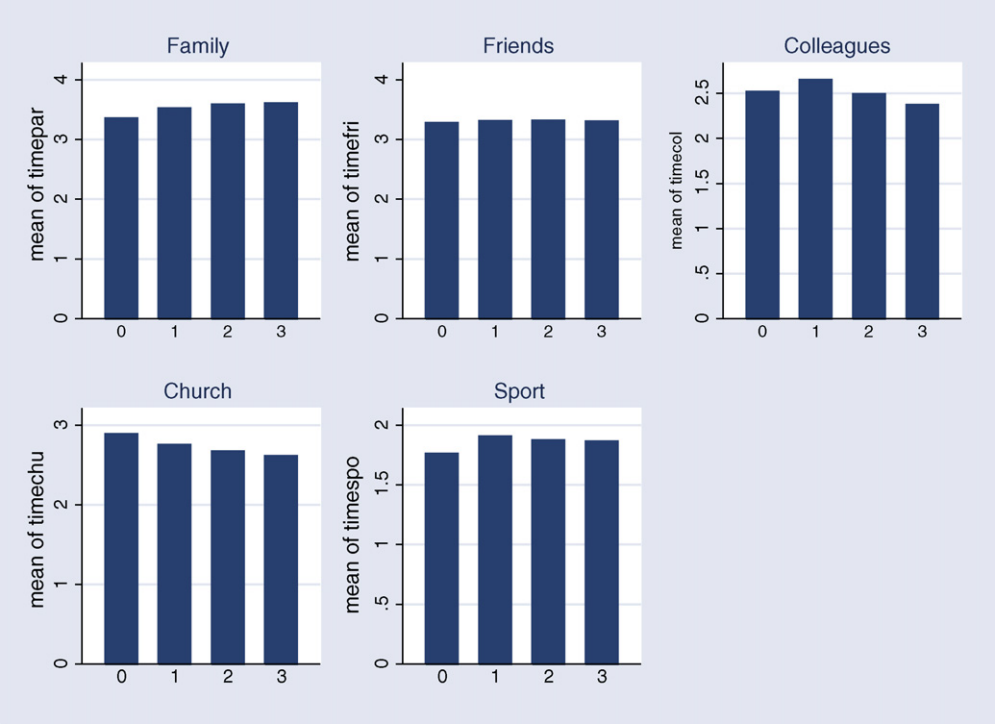

Fig. 7. TV viewing and relational activities.

In order to assess the net impact of the relational component of voluntary activities, Table 4 also reports results obtained by including both membership and active participation dummy variables, either individually (column 3) or jointly (column 4). These results are particularly interesting, as the coefficients for active involvement, net of simple membership, can be given a causal interpretation, thus allowing us to deal, albeit indirectly, with the issue of causality that undermines the interpretation of cross-sectional regression results. With the only exception of the dummy variable for the sport-related voluntary activity, which is no longer significant, all the results presented above are robust to the inclusion of the membership dummies. Active involvement in religious, art and charity organizations has a positive and significant effect on

Table 3
Life satisfaction and number of voluntary organizations

Independent variable	(a)	(b)	(c)
Vol. org. member (number)	0.31 (5.96)		0.11 (1.65)
Vol. org. active member (number)		0.54 (7.05)	0.42 (4.17)
Adjusted R^2	0.35	0.35	0.35
Observations	73,572	73,572	73,572

Note: OLS estimates. Dependent variable: life satisfaction. *t*-Statistics in parentheses (heteroskedasticity robust standard errors). Regressors also include indicators of gender, age and education, individual country dummies and time dummies for survey waves (1990–1991, 1995–1997 and 1999–2001), as detailed in Table 2. Data source: World Values Survey 2–4 (Inglehart, 2000, 2004).

Table 4
Life satisfaction and individual voluntary activities

Independent variable	(a)	(b)	(c)	(d)
Vol. org.: church	1.41 (6.03)	1.14 (4.75)	1.16 (4.12)	0.93 (3.25)
Vol. org.: sport	0.95 (4.21)	0.54 (2.29)	0.16 (0.55)	0.28 (0.88)
Vol. org.: art	1.43 (5.69)	1.03 (3.83)	0.42 (1.25)	0.72 (2.00)
Vol. org.: union	0.16 (0.49)	−0.24 (−0.69)	−0.54 (−1.50)	−0.55 (−1.43)
Vol. org.: politics	0.14 (0.39)	−0.43 (−1.18)	−0.52 (−1.21)	−0.01 (−0.02)
Vol. org.: environment	0.65 (1.61)	−0.08 (−0.18)	−0.37 (−0.75)	0.15 (0.28)
Vol. org.: professional	0.35 (1.13)	−0.13 (−0.39)	−0.54 (−1.38)	−0.02 (−0.06)
Vol. org.: charity	1.78 (6.13)	1.58 (5.11)	1.54 (3.73)	1.49 (3.43)
Adjusted R^2	0.36	0.36	0.36	0.36
Observations	77,933	75,041	77,933	75,041

Note: OLS estimates. Dependent variable: life satisfaction. *t*-Statistics in parentheses (heteroskedasticity robust standard errors). The full set of regressors is described in Table 2. Data source: World Values Survey 2–4 (Inglehart, 2000, 2004). Columns (a) and (c): coefficients estimated individually in separate equations. Columns (b) and (d): coefficients estimated jointly in a single equation.

life satisfaction. Participation in a charity voluntary organization has the largest net effect on life satisfaction (1.5 percentage points).

It is interesting to observe that membership or active involvement in unions, political parties, environment and professional voluntary organizations are not significantly related to life satisfaction. This seems to suggest that it is the activities where intrinsic motivation plays a prominent role (such as charity, church and arts) that matter most for life satisfaction, whereas the activities that imply a joint effort for a common cause (such as unions, parties, environmental organizations) are less strongly related to subjective well-being. These results can be considered as an indication that the genuineness component of relational goods is particularly relevant for individual happiness.

Table 5 presents estimation results for the link between individual indicators of relational time and life satisfaction.[28] As above, the table reports results obtained by including the relational proxies either individually (column (a)) or jointly (column (b)). When considered individually, all indicators have positive and significant coefficients. However, when the indicators are inserted jointly, the coefficients for time spent with colleagues and with people from church are no longer significant. Time spent with the family (parents and other relatives) has the largest effect on life satisfaction in both specifications, with a coefficient ranging between 1 and 0.73. The indicators for time spent with friends and with people from service activities have positive and significant coefficients, ranging between 0.8 and 0.6. Taken together, these results indicate that the identity component of relational goods, as opposed to sociality *per se*, is a key factor for individual well-being.[29]

As noted above, however, these findings could also be interpreted as the result of reverse causality: happier individuals are more likely to contribute time and efforts to interpersonal rela-

[28] Relational time indicators are only available in wave 4, which explains the difference between about 73,000 and 35,000 observations in Tables 2–6, respectively. The sample in Tables 5 and 6 includes the following countries: Spain, United States, Canada, Japan, Mexico, South Africa, Argentina, South Korea, Puerto Rico, Nigeria, Chile, India, Pakistan, China, Peru, Philippines, Moldova, Serbia, Montenegro, Macedonia, Bosnia-Hercegovina, Albania, Algeria, Indonesia, Iran, Jordan, Morocco, Vietnam, Zimbabwe, Uganda and Egypt.

[29] In order to take into account the ordinal nature of the original variables, columns (c) and (d) report results obtained when using the relational indicators defined as dummy variables. All the results discussed above are robust to the use of this alternative definition of the relational indicators.

Table 5
Life satisfaction and relational activities: OLS

Independent variable	(a)	(b)	(c)	(d)
Time spent with: family	1.00 (6.76)	0.73 (4.73)	2.10 (6.47)	1.56 (4.59)
Time spent with: friends	0.78 (5.17)	0.57 (3.51)	1.30 (4.10)	0.79 (2.31)
Time spent with: colleagues	0.27 (2.54)	0.03 (0.23)	0.76 (3.02)	0.37 (1.37)
Time spent with: church	0.24 (2.06)	0.08 (0.67)	0.56 (2.07)	0.26 (0.92)
Time spent with: service org.	0.79 (6.83)	0.62 (5.12)	1.69 (6.28)	1.34 (4.80)
Adjusted R^2	0.29	0.30	0.29	0.30
Observations	35,050	32,642	35,050	32,642

Note: OLS estimates. Dependent variable: life satisfaction. *t*-Statistics in parentheses (heteroskedasticity robust standard errors). The full set of regressors is described in Table 2. Data source: World Values Survey 4 (Inglehart, 2004). Columns (a) and (c): coefficients estimated individually in separate equations. Columns (b) and (d): coefficients estimated jointly in a single equation.

Table 6
Life satisfaction and relational activities: 2SLS

Independent variable	Coeff.	Endog.	*p*-Value	Ov. R.	*p*-Value
Time spent with: family	9.13 (3.02)	7.99	0.00	0.92	0.34
Time spent with: friends	1.34 (3.12)	1.97	0.16	1.13	0.29
Time spent with: colleagues	4.37 (3.39)	10.59	0.00	0.44	0.51
Time spent with: church	12.50 (2.54)	8.68	0.00	1.97	0.16
Time spent with: service org.	5.63 (2.90)	6.50	0.01	0.74	0.39
Observations	34,962				

Note: Dependent variable: life satisfaction. 2SLS estimates (*t*-statistics in parentheses, heteroskedasticity robust standard errors). Instruments: importance of friends and importance of family. The full set of regressors is described in Table 2. Data source: World Values Survey 4 (Inglehart, 2004).

tions. In order to deal with the simultaneous nature of the relationship between relationality and life satisfaction, Table 6 reports 2SLS results, using "importance of friends" and "importance of family" as instruments for relational time indicators. The coefficients for all indicators are positive and significant, indicating that the effect of relational time on life satisfaction is not spurious. The results of Hausman-type tests (columns 3 and 4) indicate that endogeneity is indeed a problem in estimating Eq. (1). Over-identifying restrictions tests (columns 5 and 6) do not lead us to reject the hypothesis of instrument validity. Overall, the results of instrumental variable estimation seem to support the causal interpretation that consumption of relational goods has a positive impact on life satisfaction.

5.2. Television and relational goods

We now consider the evidence for the relationship between television viewing and relational activities. Table 7 presents probit estimation results for Eq. (2), using either membership (row 1) or active participation (row 2) in voluntary organizations as the dependent variable.[30] The TV

[30] Data on television viewing are not available for all countries, which explains the fall to about 49,000 observations in Tables 7–9 for the following countries: Germany, Spain, United States, Mexico, South Africa, Australia, Norway, Sweden, Finland, Argentina, Switzerland, Puerto Rico, Brazil, Nigeria, Chile, Belarus, India, Taiwan, Lithuania, Latvia, Estonia, Ukraine, Russia, Peru, Venezuela, Uruguay, Moldova, Georgia, Armenia, Azerbaijan, Dominican Republic, Bangladesh, Serbia, Montenegro, Croatia, Bosnia-Hercegovina, Zimbabwe and Uganda.

Table 7
Television and participation in voluntary organizations

Dependent variable	TV > 2	TV > 3	TV > 3 net
Voluntary org. dummy	−0.07 (−5.05)	−0.08 (−4.92)	−0.05 (−2.76)
Active voluntary org. dummy	−0.09 (−6.55)	−0.08 (−4.98)	−0.04 (−2.01)
Pseudo R^2	0.23	0.23	0.23
Observations	46,302	46,302	46,302

Note: Probit estimates. Dependent variable in column 1. z-Values in parentheses (heteroskedasticity robust standard errors). The full set of regressors is described in Table 2. TV > 2 dummy = 1 if watch TV more than 2 h per day. TV > 3 dummy = 1 if watch TV more than 3 h per day. Data source: World Values Survey 3 and 4 (Inglehart, 2000, 2004).

Table 8
Television and number of voluntary organizations

Dependent variable	TV > 2	TV > 3	TV > 3 net
Number of voluntary org.	−0.09 (−5.72)	−0.10 (−5.48)	−0.06 (−2.85)
Number of active voluntary org.	−0.07 (−6.52)	−0.07 (−6.10)	−0.04 (−2.80)
Adjusted R^2	0.23	0.23	0.23
Observations	46,302	46,302	46,302

Note: OLS estimates. Dependent variable in column 1. t-Statistics in parentheses (heteroskedasticity robust standard errors). The full set of regressors is described in Table 2. TV > 2 dummy = 1 if watch TV more than 2 h per day. TV > 3 dummy = 1 if watch TV more than 3 h per day. Data source: World Values Survey 3 and 4 (Inglehart, 2000, 2004).

indicator is either a dummy for high-viewers (more than 2 h per day, column 2), or a dummy for heavy-viewers (more than 3 h per day, column 3), controlling for the same set of socio-economic and demographic variables as in Eq. (1). The results indicate that higher television consumption levels are associated to a lower probability of participating in voluntary organizations. Interestingly, the crowding out effect of television is larger and more significant for active participation than for pure membership. In column 4, we report the coefficients of the dummy for very heavy users when inserting both television dummy variables, so that the estimates capture the net effect of heavy viewing. The net effect is negative and significant for both pure membership and active participation. As a robustness check, Table 8 presents OLS estimation results for the same equations as above, using the number of voluntary organizations as the dependent variable. All the results are qualitatively unchanged in this alternative specification.

Table 9 presents probit estimation results for the relationship between TV viewing and participation in individual voluntary organizations. We find that both the high and the heavy viewing dummies are associated with significantly lower probabilities of active participation in religious, environment and professional voluntary organizations. Active participation to charities is also negatively affected by television viewing, although the effect is only significant for heavy viewing. When considering the net effect of heavy viewing (column 4), the same activities are negatively affected, although the effect on participation in religious and charity organizations is only marginally significant.

Table 10 presents OLS estimation results for the relationship between television viewing and individual relational activities.[31] High levels of television consumption have a negative and sig-

[31] Since relational time is only available for WVS wave 4 and data on television viewing are not available for all countries, the sample includes about 17,000 observations for the following countries: Spain, South Africa, Argentina, Puerto Rico, Nigeria, Pakistan, Peru, Bangladesh, Algeria, Indonesia, Jordan, Morocco, Zimbabwe, Uganda and Egypt.

Table 9
Television and participation in individual voluntary organizations

Dependent variable	TV > 2	TV > 3	TV > 3 net
Voluntary org.: church	−0.14 (−8.61)	−0.12 (−5.73)	−0.04 (−1.56)
Voluntary org.: sport	0.05 (2.77)	0.02 (1.04)	−0.01 (−0.38)
Voluntary org.: art	−0.03 (−1.87)	−0.02 (−1.12)	−0.01 (−0.21)
Voluntary org.: union	0.00 (0.09)	−0.02 (−0.61)	−0.02 (−0.75)
Voluntary org.: politics	−0.05 (−2.24)	0.01 (0.27)	0.05 (1.62)
Voluntary org.: environment	−0.08 (−3.30)	−0.11 (−3.54)	−0.08 (−2.19)
Voluntary org.: professional	−0.07 (−3.46)	−0.11 (−4.19)	−0.09 (−2.88)
Voluntary org.: charity	−0.03 (−1.61)	−0.05 (−2.12)	−0.04 (−1.53)
Adjusted R^2	0.14	0.14	0.14
Observations	49,994	49,994	49,994

Note: Probit estimates. Dependent variable in column 1. z-Values in parentheses (heteroskedasticity robust standard errors). The full set of regressors is described in Table 2. TV > 2 dummy = 1 if watch TV more than 2 h per day. TV > 3 dummy = 1 if watch TV more than 3 h per day. Data source: World Values Survey 3 and 4 (Inglehart, 2000, 2004).

Table 10
Television and relational time: OLS

Dependent variable	TV > 2	TV > 3	TV > 3 net
Time spent (family)	0.04 (3.41)	0.01 (0.71)	−0.02 (−1.30)
Time spent (friends)	−0.01 (−1.06)	−0.04 (−2.26)	−0.04 (−2.04)
Time spent (colleagues)	−0.14 (−7.42)	−0.13 (−5.70)	−0.05 (−2.04)
Time spent (church)	−0.05 (−2.82)	−0.04 (−1.99)	−0.01 (−0.59)
Time spent (service org.)	0.00 (0.20)	−0.01 (−0.45)	−0.02 (−0.65)
Adjusted R^2	0.19	0.19	0.19
Observations	17,125	17,125	17,125

Note: OLS estimates. Dependent variable in column 1. t-Statistics in parentheses (heteroskedasticity robust standard errors). The full set of regressors is described in Table 2. TV > 2 dummy = 1 if watch TV more than 2 h per day. TV > 3 dummy = 1 if watch TV more than 3 h per day. Data source: World Values Survey 4 (Inglehart, 2004).

nificant effect on the time spent with friends, colleagues and people from the church. The causal interpretation is reinforced by the finding that the net effect of heavy viewing (column 4) is also negative and significant in the equations for time spent with friends and colleagues.[32] It is interesting to observe that the high-television dummy has a positive and significant coefficient in the equation for time spent with the family. This is consistent with the view, shared by some observers, that television may lead to more interaction within the family because it provides one of the few opportunities for family members to spend time together (Riley et al., 1949).

As in the previous case, it should be noted that the relationship between television viewing and relationality could be simultaneous, given that people might use television as a response to solitude and, more generally, to negative affective experiences. Table 11 presents instrumental variable estimates (2SLS) for Eq. (2), using the variable "importance of television" as an instrument for TV consumption, under the assumption that self-reported television importance is related to the consumption of television, but is not independently related to relational activities. It should be observed, however, that there is a relatively small overlap in the observations for TV use (to

[32] Qualitatively similar results are obtained estimating an ordered probit model to take into account the categorical nature of the dependent variable.

Table 11
Television and relational time: 2SLS

Dependent variable	TV	Endog.	*p*-Value	Ov. R.	*p*-Value
Time spent (family)	0.02(0.28)	0.16	0.69	3.43	0.06
Time spent (friends)	−0.11(−1.74)	3.78	0.05	6.76	0.01
Time spent (colleagues)	−0.01(−0.08)	1.18	0.28	7.58	0.01
Time spent (church)	−0.23(−2.69)	4.54	0.03	9.13	0.00
Time spent (service org.)	0.06(0.72)	0.91	0.34	1.02	0.31
Observations	9325				

Note: Dependent variable in column 1. 2SLS estimates (*t*-statistics in parentheses, heteroskedasticity robust standard errors). Instruments: importance of television (categorical and dummy). The full set of regressors is described in Table 2. Data source: World Values Survey 4 (Inglehart, 2004).

be instrumented) and TV importance (the instrument). This implies that the resulting sample is relatively small (about 9300 observations).[33]

The coefficients estimated by 2SLS are qualitatively similar, in terms of both sign and size, to those obtained by OLS. However, they are estimated less precisely, so that the negative effect of television is only significant in the equations for time spent with people from the church and, marginally, time spent with friends. The results of Hausman-type endogeneity tests (columns 3 and 4) indicate that the absence of simultaneity cannot be rejected except for two equations (time spent with friends and people from church). However, over-identifying restrictions tests (columns 5 and 6) lead to rejecting the hypothesis that the instruments are valid in all equations except those for service organization and (marginally) family. Overall, these findings can be considered consistent with the causal interpretation of the link between television and relational goods.

6. Discussion

This paper presented empirical evidence from the World Values Survey on the relationship between relational goods and individual well-being. We found that the relational component of volunteering (i.e. active participation in voluntary organizations) is positively and significantly associated with higher life satisfaction, with an effect that is quantitatively similar to that of moving up by one decile in the income scale. The voluntary activities where the dimension of genuineness (non-instrumentality) is stronger, such as charity, church and art-related activities, matter most for life satisfaction. Time spent in relational activities also has a large positive effect on life satisfaction. In particular, time spent with parents and relatives has the largest effect on life satisfaction and time spent with friends and people from service organizations also has a positive and significant effect. Interestingly, relational activities where the identity dimension is less pronounced, such as time spent with people from work or church, are not significantly related to life satisfaction.

We also found a negative relationship between television viewing and consumption of relational goods. High levels of television viewing have a significant negative effect on volunteering activities and on the time spent with friends, colleagues and people from the church. Both empirical relationships are strongly significant and robust to estimation by instrumental variables to deal with possible simultaneity.

[33] The sample includes only WVS wave 4, with the following countries: Nigeria, Bangladesh, Algeria, Indonesia, Jordan, Zimbabwe, Uganda and Egypt.

Relational goods and the crowding-out effect of television can help us to answer a question that is central in the debate on the paradoxes of happiness: why do rational people allocate their time and resources without maximizing their well-being? Explanations based on *positional* or *aspiration* treadmill effects provide a consistent interpretation of such behaviour. Positional theories point to the presence of externalities that lead to excessive consumption of conspicuous goods. Aspiration theories assume that in choosing their consumption levels people do not take into account the effects on aspirations, which in turn negatively affect well-being.

A relational theory of happiness would explain the income–happiness paradox by arguing that higher income levels are associated with a tendency to over-consume material goods and under-consume relational goods, an important determinant of subjective happiness. The recent literature in social sciences offers many indications that the time devoted to interpersonal relations is falling, crowded-out by the extension of markets to domains covered in the past by non-market institutions such as family, church and civil society.[34]

In particular, the expansion of the market and the erosion of 'spaces' for interpersonal relations has led, as by-products, to greater mobility between jobs and personal areas, shift of care of children and the old from family to market (Gui and Sugden). The time and effort devoted to production and consumption of relational goods are indeed much lower today than in the past.[35] However, when material needs have been satisfied to a large degree, as it is the case in developed economies, "well-being depends to an increasing extent upon social factors, like social environment, individual relative position and social status and the ability to construct and enjoy meaningful and satisfactory relations with other people" (Antoci et al., 2005, p. 2).

If less relationality leads to less happiness, the key question is why people consume lower and lower levels of relational goods. One possible explanation comes from recent studies such as Bartolini (2006) and Antoci et al. (2005). The common idea in these works is the focus on relational goods as *public* goods: the level of relationality can be too low because of a coordination failure in contributing to the public good. In this approach, changes of relative prices over time tend to increase the costs of contributing to the production of relationality.

Scitovsky (1976) goes in a similar direction in distinguishing between "comfort" and "stimulation" goods. Comfort goods provide immediate rewards, with strongly decreasing marginal utility and tend to lead quickly to boredom. Stimulation goods, such as cultural goods, have opposite characteristics, as their marginal utility is generally increasing. Relational goods can be interpreted as stimulation goods. Indeed, as pointed out by Hirschman (1996), relational goods confer *at the same time* both comfort and stimulation. People tend to consume too much comfort goods also because they often appear as *stimulation goods in disguise*, offered at a much lower price than actual stimulation goods. In our view, television provides inexpensive and effortless artificial relationality that is often used as a substitute for actual interpersonal relationships. Television viewing can therefore be seen as a comfort good that *crowds out* consumption of relational goods.

The evidence presented in this paper suggests that the *relational treadmill* can provide an additional explanation of the income–happiness paradox. As a society becomes more affluent, the effect of higher income on individual happiness tends to be offset by lower consumption of relational goods. The pervasive and increasing role of television, through its crowding-out effect

[34] See, among others, Putnam (1995), Lane (2000) and Easterlin (2005).

[35] Ryff and Singer (2000) report evidence showing that a reduction in social relationships and social bonds (in particular, the three Aristotelian basic relational goods: friendship, love and civil participation) is strongly related to a worse self-assessment of both health and happiness.

on relational activities, helps to explain why individuals tend to consume sub-optimal levels of relationality.

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